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ENCODING METHOD AND DECODING METHOD FOR A MULTIPLE LINK COMMUNICATION NETWORK AND ASSOCIATED DEVICES

Abstract

A computer-implemented encoding method for encoding an input data stream for transmission from a transmitter coding point using a plurality of data links to improve communication networks. After segmenting the input data stream into data symbols, a link set custom-character of transmission data links is randomly selected from all data links. The data symbols are encoded by network coding. The encoded data symbol is packetized into a data packet together with decoding information and the randomly selected data links. The data packets are sent each along one randomly selected data link.

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Background/Summary

TECHNICAL FIELD

[0001] The disclosure herein relates to a computer-implemented encoding method and a complementary decoding method. The disclosure herein further relates to coding points and a communication network carrying out these methods. The disclosure herein still further relates to a computer program and a computer-readable storage medium.

BACKGROUND

[0002] Network coding is a field that has been widely investigated by the scientific communication technology community. An example for network coding is disclosed in J. K. Sundararajan, D. Shah, M. Medard, M. Mitzenmacher and J. Barros, "Network Coding Meets TCP," *IEEE INFOCOM* 2009, Rio de Janeiro, Brazil, 2009, pp. 280-288, doi: 10.1109/INFCOM.2009.5061931.

[0003] A special form of network coding, called random linear network coding (RLNC), is seen as most appropriate for actual implementation, due to its simplicity that allows efficient implementation. First commercial implementations for RLNC exist, for example the KODO library developed by Steinwurf ApS (https://www.steinwurf.com/kodo).

[0004] Network coding is done between devices called coding points, which are located on the sender, the receiver, or on a networking element in-between. Each coding point either encodes, decodes or recodes a data stream.

[0005] Encoded data may be distributed across several links towards the decoding point. In the closest related setup, network coding is implemented in an end-to-end fashion on the sender and receiver side, which are both connected via multiple parallel links, to allow more reliable transmissions.

[0006] However, network encoded links need to be configured to fit the exactly used setup, if it should be fully leveraged. Network coding parameters include the links on which the encoded data should be transmitted, coding symbol size and coding rate, among others. These parameters are typically statically assigned at the beginning of a communication session, or even in general. This results in a static setup that cannot adapt to changing link conditions.

[0007] Network coding is a sub-field to communication technology in which data is encoded on the network layer to enhance user throughput or communication reliability. To do so, the data streams are encoded within the network elements (routers, switches, middle-boxes) to create extra redundancy or increase the overall information transported in the network. Network coding elements can be placed in small-scale networks (as are present for example within aircrafts) or over-the-top in a wide-area network such as the internet.

SUMMARY

[0008] It is the object of the disclosure herein to improve network communication between coding points.

[0009] The object is achieved by the subject-matter and preferred embodiments disclosed herein.

[0010] The disclosure herein provides a computer-implemented encoding method for encoding an input data stream for transmission from a transmitter coding point through a communication network having a plurality of data links to a receiver coding point, the method comprising: [0011] segmenting the input data stream into a plurality of data symbols; [0012] randomly selecting a link set of transmission data links from all data links; [0013] generating configuration data based on the link set obtained in step b); [0014] encoding the data symbols by network coding based on the configuration data to obtain encoded data symbols and adding decoding information to the configuration data to generate control data, wherein the decoding information allows decoding of the encoded data symbols at the receiver coding point; and [0015] packetizing each encoded data symbol together with the control data into a data packet, wherein the control data causes the data packet to follow any of the transmission data links.

[0016] Preferably, in step b) the link set is randomly selected by assigning a drawing probability to each data link and by a random number generator generating a random number for each data link. Preferably, the respective data link is added to the link set, if the generated random number is smaller than or equal to the drawing probability assigned to that data link.

[0017] Preferably, the drawing probability is predetermined, preferable to be equal for all data links. Preferably, the drawing probability is set dynamically through an API. Preferably, the drawing probability for a particular data link is determined based on at least one link property of that data link. Preferably, the link property is indicative of the data packet loss, and the drawing probability increases with decreasing packet loss and vice versa. Preferably, the link properties such as link congestion, cost per data rates, etc.

[0018] Preferably, the drawing probability is chosen from an interval from 0 to 1, where endpoint 0, endpoint 1, or endpoints 0 and 1 are excluded.

[0019] Preferably, in step c) the configuration data includes any of an input symbol size, that is indicative of the number of symbols used as input for network coding; a coding rate, that is indicative of the fraction of the number of input symbols per the number of output symbols; and an output symbol size, that is indicative of the number of symbols output from the network coding.

[0020] Preferably, in step d) encoding is carried out by random linear network coding and/or by a codebook based method.

[0021] Preferably, the steps b) to e) are carried out asynchronously. Preferably, the steps b) to e) are carried out sequentially in the order b), c), d), e).

[0022] The disclosure herein provides a transmitting method for transmitting an input data stream from a transmitter coding point through a communication network having a plurality of data links to a receiver coding point, the method comprising: [0023] carrying out a preferred encoding method to obtain a data packet; [0024] with a transmitter, transmitting the data packet.

[0025] The disclosure herein provides a computer-implemented decoding method for decoding data packets received at a receiver coding point through a communication network having a plurality of data links to a transmitter coding point from that transmitter coding point, the method comprising: [0026] de-packetizing each received data packet to obtain an included encoded data symbol and control data; [0027] extracting configuration data and decoding information from the control data; and [0028] decoding the encoded data symbol into a data symbol based on the configuration data and the decoding information.

[0029] Preferably, before step a) a preferred encoding method or a preferred transmitting method according is carried out to obtain the data packets.

[0030] Preferably, step b) includes waiting for a predetermined waiting period for a number of encoded data symbols required for decoding or waiting for a maximum waiting period before commencing with step c).

[0031] The disclosure herein provides a transmitter coding point comprising means for carrying out a preferred encoding method or a preferred transmitting method.

[0032] The disclosure herein provides a receiver coding point comprising means for carrying out a preferred decoding method.

[0033] The disclosure herein provides a communication network comprising at least one transmitter coding point and at least one receiver coding point, wherein the transmitter coding point and the receiver coding point are connected through a plurality of data links.

[0034] The disclosure herein provides a computer program comprising instructions which, when executed by a suitable configured coding point, cause the coding point to carry out a preferred method.

[0035] The disclosure herein provides a computer-readable storage medium comprising the computer program.

[0036] A coding point is considered that can change its parameters, i.e. its configuration, dynamically. It is an idea to split a data stream into independent symbols that are each encoded differently. For each symbol, a set of transmission links can be chosen for transmission according to a probability distribution and then the symbol is encoded based on the considered links. By changing the probability with which a link is chosen for transmission, the traffic can be dynamically balanced across different links, in a more fine-granular and adaptive fashion than would be possible otherwise.

[0037] An advantage of the disclosed approach is that the coding point becomes configurable in a more fine-granular fashion. It can be adapted to changing link properties and to changing cost per data rates on each link and can therefore reduce the amount of transmitted data or transmission cost. Mathematically, it can be shown that by choosing the involved links randomly for each symbol, a best configuration can be calculated more easily, as the related problems turn from a combinatorial/integer problem into a continuous one. The coding point can also run in a “legacy mode”, in which certain links are chosen with 100% probability, which generates a static setup as typically used in the state of the art.

[0038] In a general setup at least two coding points are connected via a plurality of N logical data links. The logical links can be abstracted from the lower-layer data transport, i.e., the data on each link can travel over multiple communication hops, each of which can use a different technology (wired ethernet/optical, WiFi, LTE, Satcom, etc.).

[0039] The paths used by each data link may, but don't have to, be disjoint. The transmitter coding point (TX) takes a data stream as input, that may originate from an application on socket-level, or from a data stream aggregated from one or several input ports, e.g., if the coding point resides on a networking hardware (switch/router). The TX coding point buffers the input stream, segments the data into symbols and applies a network code on them, for example a random linear network code (RLNC).

[0040] In the encoding process the coding point recombines K input symbols to create L encoded symbols, where L>K. It is then said that the code has a coding rate of R=K/L. The encoded symbols are transmitted over the different links, where some of them may be lost or corrupted and dropped.

[0041] The receiver coding point (RX) receives a subset of the encoded symbols, as some of them may be lost, and applies a decoding algorithm on the received symbols to recover the original data. Recovery is (depending on the code) successful when K out of L symbols are received. The recovered data is then forwarded to the output port, which could again be a software socket or an output on the networking hardware.

[0042] The encoder CP consists of a random link decision (RLD) routine, a configurator routine (CON), an encoding routine (ENC) and a packetizer routine (PACK). In the disclosure herein, not all outgoing links have to be used all the time. The RLD can regularly draw the links to use for each encoding cycle randomly according to a given routine. In the simplest case, the RLD is given probabilities p.sub.i, where i is an index by which all links are labeled, and decides to draw each link independently according to its probability. The decision routine, or the probabilities, may be predetermined, or may be dynamically adapted via an application programming interface (API).

[0043] Once the decision on the links has been made, a configuration is determined by the CON routine. In some embodiments, the configuration consists of a field size, a generation size K and a coding rate R (note that the output symbol number L=K/R by definition of R). However, in some realizations the CON may also determine the used coding algorithm itself, the underlying mathematical field, which defines the exact form of the encoding operations, or similar.

[0044] The ENC routine accepts the configuration, takes a set of symbols from the input and produces a set of encoded symbols, as defined by the configuration. The encoded symbols are handed to the PACK routine, which creates network packets from the encoded symbols. The packets may include control information such as the target address of the packet, the used codebook for encoding, identifiers for the symbols included in the packet, or similar. Packing will

follow a predefined packet format, e.g., it could follow the one proposed by J. Heide et al., “Random Linear Network Coding (RLNC)-Based Symbol Representation” obtainable at https://datatracker.ietf.org/doc/html/draft-heide-nwcrg-rlnc-00 the contents of which are incorporated herein by reference, specifically Section 3—Symbol Representation.
[0045] Packing also may include load balancing, i.e., deciding how many of the symbols are sent over each link. Therefore, the PACK routine requires input from the RLD to know which links are used, as well as from the CON routine, to create the right control information. Finally, the PACK routine forwards the created packets to the correct output ports.
[0046] The decoder CP comprises the counterparts of the encoder, namely a de-packetizer routine (DE-PACK), a configuration extraction routine (CON-E) and a decoder routine (DEC).
[0047] The DE-PACK receives the packets sent by the PACK routine of the encoding CP and separates the control information from the encoded symbols. Both the control information and the encoded symbols are handed over to the CON-E.
[0048] The CON-E routine takes the control information and buffers the encoded symbols and determines from the control information which configuration was used. Once sufficiently many encoded symbols are present, the CON-E passes the configuration and encoded symbols to the DEC.
[0049] The DEC uses the configuration and the encoded symbols to decode the data, which is finally forwarded out of the decoder.
[0050] Finally, the API can be used by an optimization routine (OPT). The OPT routine may take external knowledge (packet losses, link rates, delays, etc.) and may adapt the behavior of the RLD routine (e.g. by changing the link probabilities p.sub.i) when necessary or on a regular basis.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0051] Embodiments of the disclosure herein are described in more detail with reference to the accompanying schematic drawings that are listed below:
[0052] FIG. **1** depicts an embodiment of a communication network;
[0053] FIG. **2** depicts an embodiment of a transmitter coding point; and
[0054] FIG. **3** depicts an embodiment of a receiver coding point.
DETAILED DESCRIPTION
[0055] Referring to FIG. **1**, an embodiment of a communication network **10** is illustrated. The communication network **10** comprises a plurality of coding points **12**. At least one coding point **12** is configured as a transmitter (TX) coding point **14** and at least one coding points **12** is configured as a receiver (RX) coding point **16**. The coding points **12** are connected through a plurality of data links **18**. Each data link **18** can be uniquely indexed, e.g., with a number i=1, 2, . . . , N.
[0056] Referring to FIG. **1** and FIG. **2**, an embodiment of the operation of the transmitter coding point **14** is described in more detail. The transmitter coding point **14** comprises a random link decision (RLD) routine **20**, a configuration (CON) routine **22**, an encoding (ENC) routine **24**, and a packetizer (PACK) routine **26**.
[0057] A data stream **28** is fed to the transmitter coding point **14**. The transmitter coding point **14**, e.g., the ENC routine **24** segments the data stream **28** into a plurality of data symbols **30**.
[0058] The ENC routine **24** is configured for network coding by a random linear network coding (RLNC) code over a finite field of size 2.sup.8 (i.e., the finite field FF(2.sup.8)). Each byte of data can be interpreted as a polynomial of eighth degree. Addition, multiplication and division operations are performed according to the binary extension field
x.sup.8+x.sup.4+x.sup.3+x.sup.2+1.
[0059] In this embodiment, the transmitter coding point **14** is connected through four independent

data links **18** to the receiver coding point **16**. Each data link **18** has a predetermined drawing probability of p.sub.1=p.sub.2=p.sub.3=p.sub.4=0.5. The RLNC code, field size and probabilities are predetermined and static. In other words, these parameters are not dynamically changed during operation.

[0060] The CON routine **22** determines a required number K of input data symbols **32**. The number K is also known as generation size K and is indicative of the number of data symbols **30** that are needed as input data symbols **32** before generating a number of L output data symbols **34**. The CON routine **22** may choose L strictly greater than K or a coding rate R from 0 to 1. The coding rate R is defined as R:=K/L.

[0061] The RLD routine **20** generates a random number for each of the data links **18**. The random number is generated from a uniform distribution out of the interval [0, 1]. The RLD routine **20** may generated the random number by a pseudo-random number generator or a true random number generator, that is typically implemented as a hardware addition.

[0062] If the RLD routine **20** determines the random number to be lower than or equal to the drawing probability p.sub.i of the respective data link **18**, that data link **18** is added to a link set custom-character of transmission data links. Otherwise, the respective data link **18** is disregarded for this cycle. The link set custom-character is passed to the CON routine **22**.

[0063] The CON routine **22** takes the link set custom-character and depending on the size |custom-character| of the link set custom-character, configuration data **36** is created in the following fashion: [0064] If |custom-character|≤2, the CON routine **22** sets K=|custom-character| and R=1, [0065] If |custom-character|≥3, the CON routine **22** sets K=2 and R=2/|custom-character|.

[0066] These conditions can be summarized by setting K=min{|custom-character|, 2} and R=K/|custom-character|. The CON routine **22** now passes the number K of input data symbols **32** and the coding rate R to the ENC routine **24**.

[0067] The ENC routine **24** accepts (K, R) from the CON routine **22** and generally waits for K input data symbols **32** to be passed from the input buffer and creates L=K/R=|custom-character| encoded data symbols **38** from it. The encoded data symbols **38** and the necessary decoding information are passed to the PACK routine **26**. The decoding information is necessary, but also sufficient, to decode the encoded data symbols **38** at the receiver coding point **16**. The decoding information, in the case of RLNC, includes the coding coefficients that were used to generate the encoded data symbols **38**.

[0068] The PACK routine **26** takes the L encoded data symbols **38**, the decoding information (e.g., the coding coefficients), the link set custom-character and the parameters (K,R) as input. The PACK routine **26** uses the coding information and (K, R) to create control data for the receiver coding point **16**.

[0069] Also, the PACK routine **26** packs each encoded data symbol **38** into an own data packet **40**. Each data packet **40** is transmitted through one of the transmission data links in the link set custom-character, preferably in a way that each transmission data link also carries exactly one data packet **40**.

[0070] Depending on the implementation, these can be carried out in a sequential manner. In the sequential operation, the RLD routine **20** chooses a link set custom-character whenever the PACK routine **26** is finished with the previous encoding cycle, and then re-run the steps.

[0071] It is also possible for each routine to run asynchronously. Asynchronous operation means that the RLD routine **20** draws for each and all data links **18** in regular predetermined time intervals, that the CON routine **22** updates the configuration data whenever a link set custom-character has been passed, and correspondingly so do the ENC routine **24** and the PACK routine **26**.

[0072] The transmitter coding point **14** may include an optimizer (OPT) routine **48**. The OPT routine **48** may measure different link properties of each data link **30**. The link properties are typically indicative of the reliability and/or quality of the data transmission along a specific data link **30**. The link properties include, for example, packet loss, cost per data rate, delay, etc. In a variation of the previously described operation, the OPT routine **48** may set the parameters, such as K, L/R, drawing probability in accordance with the link properties. In general, the drawing probability is increase, if a desired link property increases and the drawing probability is decreased if an undesired link property increases.

[0073] After the encoding is complete, the corresponding data packets **40** are transmitted to the receiver coding point **16**.

[0074] Referring to FIG. **1** and FIG. **3** an embodiment of the operation of the receiver coding point **16** is described in more detail. The receiver coding point **16** includes a depacketizer (DE-PACK) routine **42**, a configuration extraction (CON-E) routine **44**, and a decoding (DEC) routine **46**.

[0075] The DE-PACK routine **42** accepts each data packet **40**, no matter from which data link **18** it came. The DE-PACK routine **42** splits the data packet **40** into control data and encoded data symbols **38**. The control data and the encoded data symbols **38** are forwarded to the CON-E routine **44**.

[0076] The CON-E routine **44** interprets the control data and extracts the configuration data and the decoding information. Specifically, the CON-E routine **44** extracts the (K, R) and the coding coefficients for each encoded data symbol **38**. The CON-E routine **44** may wait for a certain predetermined time to receive all encoded data symbols **38** that are necessary for decoding. The wait time may have a defined maximum waiting time. The CON-E routine **44** passes the configuration data, the decoding information and the received encoded data symbols **38** to the DEC routine **46**.

[0077] The DEC routine **46** sets its configuration according to the configuration data received from the CON-E routine **44**. The DEC routine **46** accepts the encoded data symbols **38** and decoding information and attempts to decode the original data symbols **30**. If the DEC routine **46** succeeds, the decoded data symbols **30** are sent to the output port.

[0078] In the following, the description gives considerations which show in which sense the disclosed methods are able to outperform network coding according to the prior art.

[0079] As mentioned before, a key added property is that the disclosed methods are more adaptive and can better be optimized than state-of-the-art solutions, in which the outgoing links and the coding configurations remain fixed. In particular, consider for comparison a static network coding point in which the link set custom-character for outgoing links, as well as (K, R) are predetermined and fixed.

[0080] It is observed that the disclosed methods can easily be adapted to behave like a static network coding solution by setting p.sub.i=1 for all links that are in the link set custom-character of the static solution and p.sub.i=0 for those that are not such that the CON routine **22** assigns a fixed configuration (K,R) to the links.

[0081] Furthermore, the two performance parameters of throughput and symbol error rate should be considered. Here, each static configuration can be described by the triple C=(custom-character ,K,R), which is one configuration taken out of a set custom-character of possible configurations. In the static case, the configuration C remains fixed and the average throughput TP.sub.static transported over each data link **30** can be calculated from the injected throughput TP.sub.in by:

[00001] $TP_{static} = TP(C) = \text{Tpin .Math. } K/ \text{ .Math. } \mathcal{L} \text{ .Math. } = TP_{in} \text{ .Math. } R(C)$

[0082] On the other hand, the average throughput according to the disclosed disclosure herein, TP.sub.inv, is given as the average of the throughputs TP(C) that would be achieved by each configuration C=(custom-character,K,R), weighted by the probability α.sub.C that a data symbol **30** is encoded using the configuration C:

[00002] $TP_{inv} = \quad {}_{C}\, TP(C) = TP_{in} \quad {}_{C}\, R(C)$

[0083] Because according to the disclosure herein it is possible to behave like a static configuration, TP.sub.static is also one of the TP(C) on the middle term. Therefore, it can be seen

that the possible throughputs TP.sub.inv according to the disclosure herein form what is called the "convex hull" of all possible TP.sub.static. That is, TP.sub.inv can take values in between all possible TP.sub.static and therefore can be more fine granularly tuned.
[0084] Similarly, one can argue for the packet error rate. In network coding, symbol errors occur, when from a generation, not sufficiently many encoded data symbols **38** are received to decode the original data symbols **30**. This is particularly the case, when less than K out of L symbols are received, while it is not important which K symbols exactly were received.
[0085] When L symbols are transmitted over different data links **18**, there is a finite amount of combinations on which data link **18** a data symbol **30** can be lost. Let b∈{0,1}.sup.L denote a pattern describing which data symbol **30** was received, with "1" representing that it was received and "0" denoting that it was not, and let |b| denote the number of ones in b, i.e., the number of received symbols. Then, given C=(custom-character,K,R)∈custom-character, there is a probability ν.sub.b(C) that the pattern b will be observed. From this, the probability that a generation of symbols is correctly received can be calculated as:

η(*C*)=Σ.sub.|b|≥Kν.sub.b (C)
[0086] Therefore, the symbol error rate of a static configuration can be calculated as η.sub.static=η(C), with C defined as above. On the other hand, the symbol error rate of the disclosure herein is given by

η.sub.inv=custom-character
[0087] Again, it can be seen that the error rates in the disclosure herein form the convex hull of that of a static system. Finally, when coming to optimization of the configuration, the disclosure herein also has some advantages. Optimization of a fixed configuration often leads to combinatorial problems, due to the parameters custom-character, K and R being taken from discrete sets. On the other hand, in the disclosure herein the optimization goes towards optimizing the probabilities α.sub.C, which are from continuous sets. This allows the consideration as continuous problems which typically are easier to solve.
[0088] While at least one example embodiment of the invention(s) is disclosed herein, it should be understood that modifications, substitutions, and alternatives may be apparent to one of ordinary skill in the art and can be made without departing from the scope of this disclosure. This disclosure is intended to cover any adaptations or variations of the example embodiment(s). In addition, in this disclosure, the terms "comprise" or "comprising" do not exclude other elements or steps, the terms "a", "an" or "one" do not exclude a plural number, and the term "or" means either or both. Furthermore, characteristics or steps which have been described may also be used in combination with other characteristics or steps and in any order unless the disclosure or context suggests otherwise. This disclosure hereby incorporates by reference the complete disclosure of any patent or application from which it claims benefit or priority.
LIST OF REFERENCE SIGNS
[0089] **10** communication network [0090] **12** coding point [0091] **14** transmitter (TX) coding point [0092] **16** receiver (RX) coding point [0093] **18** data link [0094] **20** random link decision (RLD) routine [0095] **22** configuration (CON) routine [0096] **24** encoding (ENC) routine [0097] **26** packetizer (PACK) routine [0098] **28** data stream [0099] **30** data symbol [0100] **32** input data symbol [0101] **34** output data symbol [0102] **36** configuration data [0103] **38** encoded data symbol [0104] **40** data packet [0105] **42** depacketizer (DE-PACK) routine [0106] **44** configuration extraction (CON-E) routine [0107] **46** decoding (DEC) routine [0108] **48** optimizer (OPT) routine [0109] custom-character link set

Claims

1. A computer-implemented encoding method for encoding an input data stream for transmission from a transmitter coding point through a communication network having a plurality of data links to a receiver coding point, the method comprising: a) segmenting the input data stream into a plurality of data symbols; b) randomly selecting a link set custom-character of transmission data links from all data links; c) generating configuration data based on the link set custom-character obtained in step b); d) encoding the data symbols by network coding based on the configuration data to obtain encoded data symbols and adding decoding information to the configuration data to generate control data, wherein the decoding information allows decoding of the encoded data symbols at the receiver coding point; and e) packetizing each encoded data symbol together with the control data into a data packet, wherein the control data causes the data packet to follow any of the transmission data links.
2. The encoding method according to claim 1, wherein in step b) the link set custom-character is randomly selected by assigning a drawing probability to each data link and by a random number generator generating a random number for each data link, wherein a respective data link is added to the link set custom-character, if the generated random number is smaller than or equal to the drawing probability assigned to that data link.
3. The encoding method according to claim 2, wherein the drawing probability is predetermined, and can be equal for all data links; and/or wherein the drawing probability is set dynamically through an API; and/or wherein the drawing probability for a particular data link is determined based on at least one link property of that data link.
4. The encoding method according to claim 1, wherein in step c) the configuration data includes any of an input symbol size, that is indicative of a number of symbols used as input for network coding; a coding rate, that is indicative of a fraction of the number of input symbols per a number of output symbols; and an output symbol size, that is indicative of the number of symbols output from the network coding.
5. The encoding method according to claim 1, wherein in step d) encoding is carried out by random linear network coding and/or by a codebook based method.
6. The encoding method according to claim 1, wherein the steps b) to e) are carried out asynchronously or sequentially in an order b), c), d), e).
7. A transmitting method for transmitting an input data stream from a transmitter coding point through a communication network having a plurality of data links to a receiver coding point, the method comprising: a) carrying out an encoding method according to any of the preceding claims to obtain a data packet; and b) with a transmitter, transmitting the data packet.
8. A computer-implemented decoding method for decoding data packets received at a receiver coding point through a communication network having a plurality of data links to a transmitter coding point from that transmitter coding point, the method comprising: a) de-packetizing each received data packet to obtain an included encoded data symbol and control data; b) extracting configuration data and decoding information from the control data; and c) decoding the encoded data symbol into a data symbol based on the configuration data and the decoding information.
9. The decoding method according to claim 8, wherein before step a) an encoding method is carried out to obtain the data packets, the encoding method comprising a computer-implemented encoding method for encoding an input data stream for transmission from a transmitter coding point through a communication network having a plurality of data links to a receiver coding point, the method comprising: a) segmenting the input data stream into a plurality of data symbols; b) randomly selecting a link set custom-character of transmission data links from all data links; c) generating configuration data based on the link set custom-character obtained in step b); d) encoding the data symbols by network coding based on the configuration data to obtain encoded data symbols and adding decoding information to the configuration data to generate control data, wherein the decoding information allows decoding of the encoded data symbols at the receiver coding point;

and e) packetizing each encoded data symbol together with the control data into a data packet, wherein the control data causes the data packet to follow any of the transmission data links.

10. The decoding method according to claim 8, wherein before step a) a transmitting method is carried out to obtain the data packets, the transmitting method comprising a transmitting method for transmitting an input data stream from a transmitter coding point through a communication network having a plurality of data links to a receiver coding point, the method comprising: a) carrying out an encoding method according to any of the preceding claims to obtain a data packet; and b) with a transmitter, transmitting the data packet.

11. The decoding method according to claim 8, wherein step b) includes waiting for a predetermined waiting period for a number of encoded data symbols required for decoding or waiting for a maximum waiting period before commencing with step c).

12. A transmitter coding point for carrying out the encoding method according to claim 1.

13. A transmitter coding point for carrying out the transmitting method according to claim 7.

14. A receiver coding point for carrying out the decoding method according to claim 8.

15. A communication network comprising at least one transmitter coding point according to claim 12 and at least one receiver coding point, wherein the transmitter coding point and the receiver coding point are connected through a plurality of data links.

16. A computer program comprising instructions which, when executed by a suitable configured coding point, cause the coding point to carry out the method according to claim 1.

17. A computer-readable storage medium comprising the computer program of claim 16.